TEXORA · LECTURE SHEET

Probability

Sample spaces and events, the classical definition and the axioms, the addition and multiplication rules, independence, conditional probability, the law of total probability and Bayes theorem, with a first look at expectation.

STAT-2109 · PROBABILITY AND STATISTICS · TOPIC 6 · LEVEL 2 TERM 1

1. Random experiments, sample spaces and events

Probability is the mathematics of uncertainty. It begins with a **random experiment**: a procedure whose outcome cannot be predicted with certainty in advance, even though all possible outcomes are known. Tossing a coin, rolling a die, drawing a card, and testing a fabric specimen for its breaking load are all random experiments in this sense, because repetition under identical conditions gives varying results.

The **sample space** S is the set of all possible outcomes of the experiment. For a coin toss S = {H, T}; for a die S = {1, 2, 3, 4, 5, 6}; for two dice S contains 36 equally likely ordered pairs. An **event** is any subset of S: the event that a die shows an even number is {2, 4, 6}. An event with a single outcome is an elementary event; the empty set is the impossible event; S itself is the certain event. Events can be combined by the set operations of union (A or B), intersection (A and B) and complement (not A). Two events are **mutually exclusive** (disjoint) when they cannot occur together, that is, when $A \cap B$ is empty.

Counting the sample space correctly is half of every elementary probability problem. With two dice, for example, (2, 5) and (5, 2) are distinct outcomes, because the dice are distinguishable by position or colour. Forgetting this is the most common source of wrong answers at this level.

2. The classical definition

When all outcomes of a finite sample space are equally likely, the **classical definition** assigns to an event A the probability

$$P(A) = \text{Number of favourable outcomes} / \text{Total number of outcomes} = n(A) / n(S)$$

The definition is simple and powerful, but its limits must be understood. It applies only when the outcomes are finite in number and genuinely equally likely. A loaded die, a worn coin, or the breaking strength of a yarn specimen have no equally likely finite outcomes, so their probabilities must come from relative frequencies observed in repeated trials or from a probability model. The classical

definition is nevertheless the foundation: coin, die, card and ball problems dominate elementary work precisely because equally likely outcomes can be counted.

WORKED EXAMPLES

Two fair dice are rolled. There are 36 equally likely outcomes.

P(sum 7): the favourable pairs are (1,6), (2,5), (3,4), (4,3), (5,2), (6,1), so P = 6/36 = 1/6 ≈ **0.167**.

P(sum 10): the favourable pairs are (4,6), (5,5), (6,4), so P = 3/36 = 1/12 ≈ **0.083**.

A single card is drawn from a well shuffled pack of 52: P(ace) = 4/52 = 1/13 ≈ **0.077**.

Note the pattern in the dice example: sums near the middle have more favourable pairs, so 7 is the most likely sum, while the extreme sums 2 and 12 have only one pair each. Probability is, at this level, careful counting.

3. The axioms of probability

Modern probability rests on three axioms, due to Kolmogorov, which every definition and every rule must respect. For any event A in a sample space S:

$$\text{Axiom 1: } 0 \le P(A) \le 1 \qquad \text{Axiom 2: } P(S) = 1 \qquad \text{Axiom 3: for mutually exclusive events, } P(A \cup B) = P(A) + P(B)$$

Axiom 1 says a probability is a number between impossibility (0) and certainty (1). Axiom 2 says something in the sample space must happen. Axiom 3, finite additivity, extends by induction to any finite collection of pairwise mutually exclusive events: the probability of the union is the sum of the probabilities. Familiar consequences follow at once: P(empty set) = 0, and the complement rule P(not A) = 1 − P(A). The complement rule is a practical workhorse, because the probability of at least one success is often far easier to find as 1 minus the probability of no successes than by direct counting.

4. The addition rule

Axiom 3 already gives the addition rule for mutually exclusive events. When events can overlap, the overlap must be subtracted once, because it would otherwise be counted twice:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

This **general addition rule** holds for all events, mutually exclusive or not; when A and B are mutually exclusive, $P(A \cap B) = 0$ and the general rule reduces to the special one. A Venn diagram makes the subtraction obvious: adding the two circles counts their lens shaped overlap twice.

WORKED EXAMPLE

One card is drawn from a pack of 52. Find P(red or king). There are 26 red cards and 4 kings, but 2 cards are both red and kings, so

P(red or king) = 26/52 + 4/52 − 2/52 = 28/52 = 7/13 ≈ **0.538**.

Omitting the subtraction would give 30/52, an overcount of exactly the two red kings.

5. Conditional probability and the multiplication rule

Information changes probability. The **conditional probability** of A given that B has occurred is the probability of A within the reduced sample space where B is known to hold:

$$P(A \mid B) = P(A \cap B) / P(B) , \quad \text{provided } P(B) > 0$$

Rearranging gives the **multiplication rule**, valid for all events:

$$P(A \cap B) = P(B) \cdot P(A \mid B) = P(A) \cdot P(B \mid A)$$

The multiplication rule is the engine for sequential problems: draws from a bag, successive inspections, stages of a process. The key step in each problem is deciding whether the second probability is conditional (the situation has changed) or unchanged (it has not). Drawing **without replacement** changes the contents of the bag, so conditional probabilities are needed; drawing **with replacement** restores the bag, so the draws are independent.

WORKED EXAMPLE

A bag contains 4 red and 6 black balls. Two balls are drawn without replacement. Find P(both red).

P(first red) = 4/10. Given a red ball has gone, 3 red balls remain among 9, so P(second red | first red) = 3/9.

P(both red) = (4/10)(3/9) = 12/90 = 2/15 ≈ **0.133**.

6. Independence

Events A and B are **independent** when the occurrence of one does not alter the probability of the other: P(A | B) = P(A) and P(B | A) = P(B). Substituting into the multiplication rule gives the defining test and the working formula for independent events:

$$P(A \cap B) = P(A) \cdot P(B) \quad \text{if and only if A and B are independent}$$

Independence must not be confused with mutual exclusivity; the two ideas are almost opposites. Mutually exclusive events are strongly dependent: if A occurs, B certainly did not. Two separate

physical devices, such as a coin and a die, generate independent events because neither can influence the other.

WORKED EXAMPLE

A coin is tossed and a die is rolled. The experiments do not affect each other, so the events are independent:
P(head and 6) = P(head) × P(6) = 1/2 × 1/6 = **1/12** ≈ 0.083.

In textile testing, independence is an assumption to be justified, not a default. Two specimens cut from adjacent positions in the same fabric piece may share a local weakness and fail dependently, whereas specimens sampled from different production lots are much more plausibly independent. Stating the assumption explicitly is part of a correct solution.

7. Total probability and Bayes theorem

Suppose the sample space is divided into mutually exclusive and exhaustive causes B_1, B_2, …, B_k, and A is an effect that can arise under any of them. The **law of total probability** assembles the probability of the effect from its contributions under each cause:

$$P(A) = \Sigma\, P(B_i) \cdot P(A \mid B_i)$$

Bayes theorem then reverses the conditioning: having observed the effect A, it gives the probability that a particular cause B_i was responsible:

$$P(B_i \mid A) = P(B_i) \cdot P(A \mid B_i) / P(A)$$

The ingredients have standard names. $P(B_i)$ is the **prior** probability of the cause, $P(A \mid B_i)$ is the **likelihood** of the effect under that cause, and $P(B_i \mid A)$ is the **posterior** probability after observing the effect. Bayes theorem is the formal rule for learning from evidence, and it is the correct tool whenever a defect, a positive test result, or a failure must be traced back to its most likely source.

WORKED EXAMPLE

Machine A makes 60% of items with 2% defective; machine B makes 40% with 3% defective. An item is chosen at random.
Total probability of a defective item: P(defective) = 0.6 × 0.02 + 0.4 × 0.03 = 0.012 + 0.012 = **0.024**.
Given the item is defective, the probability it came from A is P(A | defective) = 0.012 / 0.024 = **0.50**, and P(B | defective) = 0.012 / 0.024 = **0.50**.
Although machine B has the higher defect rate, it makes fewer items, and the two contributions to

> the defective pool are exactly equal, so a defective item is equally likely to have come from either machine.

The example repays attention because intuition often fails here. Many students answer that a defective item more likely came from B because B is the sloppier machine. The calculation shows why that is wrong: what matters is each machine share of the defective pool, which combines both its production share and its defect rate.

8. Expectation: a first look

The **expected value** of a numerical random variable X is its long run average, the probability weighted mean of its possible values:

$$E(X) = \Sigma\, x \cdot P(X = x)$$

For a fair die, each face has probability 1/6, so E(X) = (1 + 2 + 3 + 4 + 5 + 6) / 6 = 21 / 6 = **3.5**. No single roll can show 3.5; the expectation is not a prediction for one trial but the average approached over many trials. Expectation links this topic to the descriptive statistics of earlier topics: the mean of a distribution and the expectation of a random variable are the same idea, once for observed data and once for a probability model. Full treatments of random variables and named distributions belong to later study; the definitions and rules of this sheet are the platform on which they stand.

Glossary

TERM	MEANING
Sample space	The set S of all possible outcomes of a random experiment.
Event	Any subset of the sample space; an outcome of interest.
Mutually exclusive	Events that cannot occur together; their intersection is empty.
Classical probability	P(A) = n(A) / n(S) when all outcomes are equally likely.
Conditional probability	P(A | B) = P(A ∩ B) / P(B): the probability of A given that B has occurred.
Independent events	Events where one occurring does not change the probability of the other, so P(A ∩ B) = P(A) P(B).
Law of total probability	P(A) = Σ P(B_i) P(A | B_i) over an exhaustive set of mutually exclusive causes.

Bayes theorem	The rule that reverses conditioning to give $P(B_i \mid A)$ from the priors and likelihoods.
Expected value	$E(X) = \Sigma\, x\, P(X = x)$, the long run average of a random variable.

Key points & summary

- Probability starts from the sample space; an event is a subset, and correct counting decides most elementary problems.
- Classical probability, $P(A) = n(A) / n(S)$, needs finite, equally likely outcomes; the axioms require $0 \le P \le 1$, $P(S) = 1$ and additivity for mutually exclusive events.
- General addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; the subtraction corrects the double counted overlap.
- Multiplication rule: $P(A \cap B) = P(B)\, P(A \mid B)$; for independent events this becomes $P(A)\, P(B)$.
- Drawing without replacement uses conditional probabilities, as in $(4/10)(3/9) = 2/15$ for two red balls; independent devices give products such as $1/2 \times 1/6 = 1/12$.
- Total probability assembles an effect from its causes, and Bayes theorem reverses it: in the machine example $P(\text{defective}) = 0.024$ and $P(A \mid \text{defective}) = 0.50$.
- Expectation is the probability weighted mean; for a fair die $E(X) = 3.5$.

Practice

Answer in your own words; brief model answers follow.

1. Two dice are rolled. Find P(sum 7) and P(sum 10), listing the favourable pairs in each case.
2. A card is drawn at random. Find P(ace) and P(red or king), explaining why a subtraction is needed in the second calculation.
3. A bag holds 4 red and 6 black balls. Two balls are drawn without replacement. Find the probability that both are red, and explain why conditional probability is required.
4. Distinguish independent events from mutually exclusive events, and use independence to find P(head and 6) when a coin is tossed and a die is rolled.
5. Machine A makes 60% of items with 2% defective and machine B makes 40% with 3% defective. Find the overall probability that an item is defective.
6. Continuing Question 5, a defective item is found. Use Bayes theorem to find the probability that it came from machine A, and explain the result in words.

Answer key

1. Sample space size 36. Sum 7: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1), so P = 6/36 = 1/6 ≈ 0.167. Sum 10: (4,6), (5,5), (6,4), so P = 3/36 = 1/12 ≈ 0.083.
2. P(ace) = 4/52 = 1/13 ≈ 0.077. P(red or king) = 26/52 + 4/52 − 2/52 = 28/52 = 7/13 ≈ 0.538. The subtraction removes the two red kings, which are counted in both the 26 red cards and the 4 kings.
3. P = (4/10)(3/9) = 12/90 = 2/15 ≈ 0.133. Conditional probability is required because removing the first ball changes the bag to 3 red balls out of 9, so the second probability depends on the first result.
4. Mutually exclusive events cannot happen together, so each occurring excludes the other; independent events do not influence each other at all. Coin and die are independent, so P(head and 6) = 1/2 × 1/6 = 1/12.
5. P(defective) = 0.6 × 0.02 + 0.4 × 0.03 = 0.012 + 0.012 = 0.024.
6. P(A | defective) = (0.6 × 0.02) / 0.024 = 0.012 / 0.024 = 0.50. Machine A contributes exactly half of all defective items, because its larger production share exactly balances its lower defect rate, so a defective item is equally likely to have come from either machine.

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